Backup slides

What Factors Are Most Associated with Football Player Market Value?

Niloy Saha · DATA 400 Individual Mini-Project · for questions

Backup A: how the dataset was cleaned

Step	Rows dropped	Rows remaining
Player-seasons in scope (14 leagues, 2020–2024)		32,966
Missing position	9	32,957
Missing date of birth	5	32,952
No valuation by July 31	490	32,462
Valuation older than 365 days	416	32,046
Zero minutes	0	32,046
Final dataset	**920**	**32,046**

23,328

rows with ≥450 minutes (72.8%) used for per-90 rates; others kept for everything else

Backup B: K-means details

k	Inertia	Silhouette
2	19,125	0.219
3	16,201	0.215
4	13,850	0.207
5	12,093	0.214
6	11,075	0.209

Cluster (k = 4)	n	Age	Min	G/90	A/90	Value
Regular high-minute starters	1,576	25.9	2,308	0.37	0.15	€8M
Creative attackers	1,018	24.6	1,234	0.28	0.35	€5M
Younger, lower output	2,046	23.0	1,029	0.18	0.10	€1.5M
Older, fewer minutes	1,455	30.0	1,124	0.23	0.10	€1M

- scikit-learn KMeans, n_init = 20, random_state = 0; StandardScaler; per-90 rates capped at the 99th percentile.
- Stability (adjusted Rand index vs the seed-0 fit): minimum 0.92, median 0.97 across 10 seeds; minimum 0.89, median 0.92 on 80% subsamples.
- Clusters account for ~23% of log-value variance among these attackers.

Backup C: rule-based segments (not clusters)

Segment	Rule (applied in order)	n	G/90	A/90	Median value
High scorers	Goals/90 ≥ 0.40 (top quarter)	1,524	0.53	0.15	€6M
Creative attackers	Assists/90 ≥ 0.23, not a high scorer	1,074	0.21	0.31	€4M
Young (under 24)	Everyone else, under 24	1,235	0.19	0.11	€2M
Prime age (24-28)	Everyone else, 24 to 28	1,456	0.19	0.10	€1.8M
Experienced (29+)	Everyone else, 29 or older	806	0.20	0.10	€900K

Defined by me with explicit cut-offs from the data; easier to explain than K-means, but capture less value variation (~14% vs ~23%).

Backup D: median market value by league



€4.5M

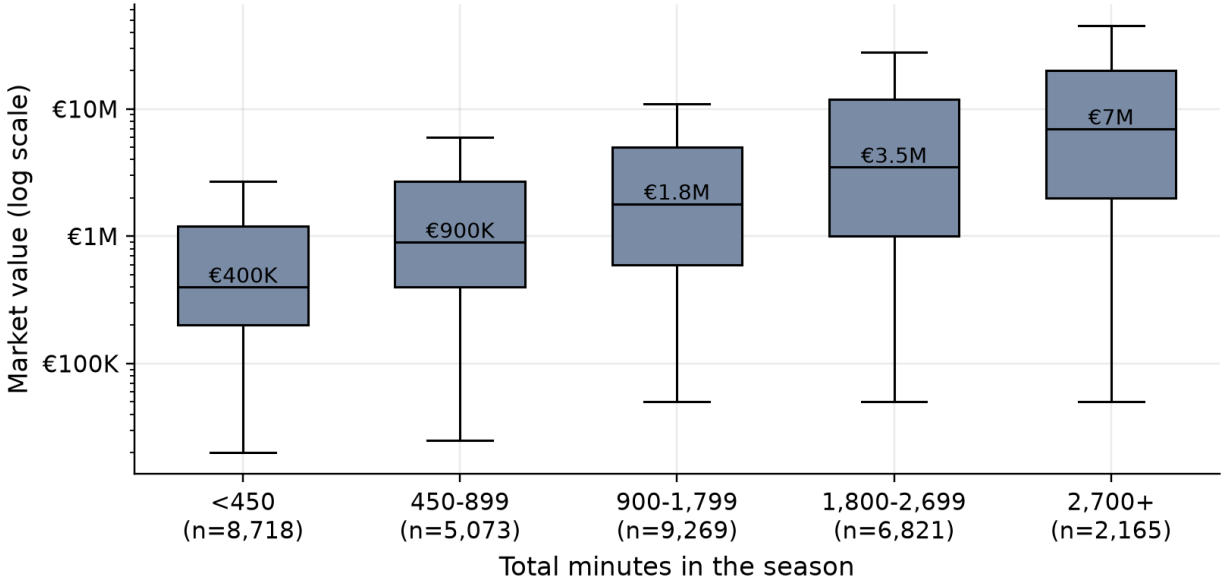
pooled median, top-five leagues

€600K

pooled median, other nine leagues

Every top-five median is €3M or more; every other league is €1M or less.

Backup E: playing time and value



0.51

Spearman, total minutes vs value (all players)

Causal direction is unclear: clubs may play players they already value, playing may raise visibility, and ability or club level may affect both. None of this was tested.